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Using Deep Learning to Optimize the Allocation of Rural Education Resources Under the Background of Rural Revitalization

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ABSTRACT

Through a large amount of literature research, the current status of rural education resource allocation, the application of deep learning in education resource optimization, and the challenges and solutions faced are sorted out. A rural education resource optimization model based on deep learning is designed, and data collection and analysis, modeling of education resource allocation optimization problems, and optimization algorithm design based on deep learning are elaborated in detail. After experimental evaluation, the optimization algorithm based on deep learning is compared with the traditional optimization method. The results show that the deep learning model has significant advantages in resource allocation accuracy, educational equity, and teaching quality improvement. The research results provide a scientific basis for education decision makers, help the rational allocation and efficient use of rural education resources, and promote the implementation of the rural revitalization strategy.

KEYWORDS

Rural Revitalization, Deep Learning, Rural Education Resources, Resource Allocation Optimization, Educational Equity

INTRODUCTION

The rural revitalization strategy is an important part of China's current development strategy. It aims to promote the comprehensive development of rural economy, culture, environment, and other aspects; improve the living standards of farmers; and promote the coordinated development of urban and rural areas. As the urban-rural gap continues to widen, the implementation of rural revitalization urgently needs the guidance of innovative thinking and technology (Guo & Li, 2024). As the cornerstone of social development, education is an indispensable and important part of rural revitalization. The level of education directly determines the ability of rural areas to cultivate talents and scientific and technological innovation; it also is the key to improving the quality of the rural population and promoting economic development (Safarov et al., 2023).

In this context, promoting the development of rural education is particularly important. Especially in today's information and intelligent world, the introduction of scientific and technological means has provided a new way to optimize educational resources. Through the application of advanced

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technologies such as deep learning, educational management departments can more accurately identify and solve problems such as uneven distribution of rural educational resources, poor teaching quality, and lack of educational facilities, thereby providing strong support for the implementation of the rural revitalization strategy. Rural revitalization is about not only economic revitalization (Zhang, 2024) but also the comprehensive improvement of social fields such as culture and education, and the optimal allocation of educational resources is the key link (Tang et al., 2024).

Although China has vigorously promoted rural education in the past few decades and achieved certain results, rural education still faces many problems overall. First, the distribution of educational resources is uneven, with a large gap in educational resources between urban and rural areas and between regions. In particular, in the western and remote rural areas, the shortage of educational resources is more serious: School facilities are old, the teaching staff is weak, and the teaching quality is difficult to guarantee. Second, there is insufficient investment in rural education, especially high-quality teaching resources (e.g., high-quality teaching materials, teaching equipment, and professional educational services) that are generally lacking (Narantuya et al., 2022). Furthermore, the degree of modernization of education in rural areas is low, the application of information-based educational resources is limited, and distance education and digital teaching have not been widely promoted in many places (Araujo et al., 2020). In addition, the teaching staff in rural areas also has the problem of high mobility and uneven teaching level. Many excellent teachers choose to leave owing to problems with treatment and working conditions, resulting in a shortage of teachers. At the same time, the educational concepts and teaching methods of some teachers are relatively backward and cannot adapt to the needs of education in the new era. Therefore, the challenge of how to optimize and allocate rural educational resources so that they can truly meet the needs of a rural revitalization strategy for talent training has become an urgent problem to be solved (Shen, 2021).

Deep learning, as an important branch of artificial intelligence, has made breakthrough progress in many fields in recent years. By simulating the working principle of a human brain neural network, deep learning can autonomously learn features, identify patterns, and make predictions and classifications from a large amount of data. In the field of education, deep learning technology has gradually penetrated into many aspects, such as personalized recommendation of educational content, evaluation of teaching effectiveness, and intelligent tutoring, providing a new technical means for the optimal allocation of educational resources (Zhang et al., 2024). Specifically, in the optimization of rural educational resources, deep learning can play an important role in the following aspects: First, it plays a role in precise allocation of educational resources. Deep learning can be used to analyze a large amount of educational data to identify the current status and demand of educational and accurately evaluate the resource shortage situation in various regions, thereby helping educational management departments to allocate resources scientifically and reasonably. Second, deep learning can be used as an intelligent teaching model. Deep learning technology can provide personalized learning plans for students in rural areas. Through intelligent teaching systems, it can analyze students' learning progress and problems, provide targeted tutoring, and improve teaching efficiency. Third, deep learning can be used for the improvement of education quality. Deep learning technology can help education departments monitor education quality in real time and evaluate and improve teaching effectiveness. For example, through technologies such as image recognition and speech analysis, students' classroom performance can be evaluated, teaching strategies can be adjusted in real time, and teaching quality can be improved (Liu & Zhu, 2024).

The advantages of deep learning are reflected not only in data analysis and processing capabilities but also in its ability to conduct autonomous learning and self-optimization. Through continuous feedback and updates, deep learning can continuously improve resource allocation plans and make dynamic adjustments based on actual conditions, thereby ensuring the efficient use of educational resources.

The main purpose of this study is to explore how to use deep learning technology to optimize the allocation of rural education resources and improve the quality and efficiency of education services.

Specifically, in this study I analyze the current situation and bottlenecks of education resources in rural areas by building an education resource allocation optimization model based on deep learning, and I then propose effective optimization solutions. Through a large amount of data collection and analysis, combined with the powerful data processing capabilities of deep learning, I aim to provide a scientific basis for education decision-makers to help achieve the rational allocation and efficient use of education resources.

RELATED RESEARCH REVIEW

Problems in the Allocation of Rural Education Resources

With the implementation of the rural revitalization strategy, the challenge of how to optimize the allocation of rural education resources has become a hot topic of academic concern. In rural areas, the shortage and uneven distribution of educational resources have become increasingly prominent, especially in remote areas, where the lack of educational resources has led to low education quality. Studies have shown that the problem of allocation of rural education resources is reflected not only in insufficient capital investment but also in the uneven distribution of educational resources in different regions, with the gap between urban and rural areas being particularly significant. In particular, in some western and mountainous areas (Zhang et al., 2023), the construction of educational facilities lags behind, and high-quality educational resources are difficult to penetrate into the grassroots, resulting in an imbalance in urban and rural education development.

Some studies have pointed out that although the national and local governments have achieved certain results in increasing investment in education funds, owing to the low level of informatization, educational resources are still unevenly distributed. To improve this situation, many scholars have proposed solutions to optimize the allocation of educational resources through digitization and informatization. By using technologies such as big data and cloud computing, educational management departments can more accurately grasp the educational resource needs of each region, thereby formulating a more scientific resource allocation strategy. These solutions provide a new perspective for solving the problem of educational resource allocation, and they also highlight the huge potential of modern technology in the allocation of educational resources (Wang et al., 2023).

The introduction of digital education has brought revolutionary changes to the allocation of educational resources, especially in the rational allocation of rural educational resources. Researchers believe that with the help of information technology, especially big data analysis, resource gaps can be effectively identified and filled. This approach not only improves the accuracy of resource allocation but also reduces the waste of resources caused by information asymmetry, helping to solve the problems that have long plagued rural education (Xue et al., 2021).

Application of Deep Learning in Educational Resource Optimization

In recent years, deep learning, as an important branch of artificial intelligence, has been widely used in many fields, and its potential in optimizing educational resources has gradually been tapped. Deep learning can help identify problems in the allocation of educational resources through deep mining and analysis of massive data, thereby providing strong data support for educational decision-making. Many studies have shown that the application of deep learning technology in the allocation of educational resources, especially in demand forecasting and resource allocation optimization, has significant advantages. In some studies, deep learning is used to process and analyze large amounts of educational data to help educational managers better understand resource use and demand changes. By establishing a deep learning model, educational management departments can accurately predict the future demand for educational resources in a certain area and then allocate resources in advance (Wang et al., 2024). This process not only improves the efficiency of resource allocation but also avoids resource waste and shortages, ensuring that educational resources are used

reasonably. At the same time, deep learning is also widely used in education quality assessment. In rural areas, because of the weak educational infrastructure and differences in teaching quality, deep learning is used to analyze students' learning behavior, classroom performance, and changes in grades. By analyzing students' learning under different teaching modes, deep learning can provide teachers with personalized teaching strategies and help optimize teaching methods. A further application is to use deep learning technology to track the quality of education in real time during the education evaluation process, promptly identify teaching problems, and make adjustments, thereby improving the quality of education (Xiao et al., 2022).

The advantage of deep learning is also reflected in its powerful self-learning ability. Based on a large amount of teaching data, the deep learning model can be continuously optimized and adjusted, gradually improving the prediction accuracy of resource allocation and educational effects. The self-iteration and optimization characteristics of this technology enable it to make more flexible and accurate decisions on resource allocation, providing an efficient and intelligent tool for educational decision-makers (Guo et al., 2020).

Challenges and Solutions in the Allocation of Rural Education Resources

Although modern technology, especially deep learning, has shown its potential in optimizing the allocation of educational resources in some studies, in practical applications, the allocation of rural educational resources still faces many challenges. One of the most prominent challenges is the lack of data and the low quality of data. In some remote areas, owing to weak technical infrastructure, collecting and processing educational data are still difficult. As a result, the allocation of educational resources in many places still relies on traditional manual methods and cannot achieve precise resource allocation. Another major challenge is the limitation of technical infrastructure. Even though advanced technologies such as deep learning can theoretically solve the problems in the allocation of educational resources, in many rural areas, information construction is still lagging behind. The lack of high-speed Internet and smart terminal devices makes it difficult to popularize the application of educational technology. For some areas with weak infrastructure, the challenge of how to improve the popularity of educational technology and provide the necessary hardware support for the application of technology is still an urgent problem to be solved (Ohta et al., 2022; Sun et al., 2024).

In response to these challenges, researchers have proposed a number of solutions. On one hand, through policy guidance, local governments can be encouraged to increase investment in rural education informatization, improve the coverage of Internet infrastructure, and gradually establish a complete education data platform. On the other hand, edge computing and other technologies can be used to process and analyze data in areas with limited resources, reduce dependence on efficient Internet, and provide support for deep learning applications in remote areas. In addition, strengthening data sharing and cooperation is also an effective solution. Integrating education data from different regions and departments enables a national education data sharing platform to be established, thereby improving the transparency and fairness of resource allocation (Lu et al., 2022; Liu, 2024).

With the development of technology, the application prospects of deep learning in the allocation of educational resources are broad. Although it faces certain challenges, with the support of policies, the improvement of infrastructure and the advancement of technology, the application of deep learning in the allocation of rural educational resources will gradually take root and become an important tool for optimizing educational resources and improving teaching quality (Wang, 2022).

DESIGN OF RURAL EDUCATION RESOURCE OPTIMIZATION MODEL BASED ON DEEP LEARNING

Data Collection and Analysis

When designing an educational resource optimization model, educational management departments first need sufficient data collection and analysis. Data-driven optimization methods rely on a large amount of accurate input data. In the next three sections I describe in detail data collection and analysis (Wei et al., 2022; Zhou, 2024).

Data Collection

The optimization of educational resources involves many factors, so it is necessary to collect multiple types of data. The main data categories include educational resource data, student demand data, and education quality data.

The educational resource data category consists of educational resources in each region or school, including the number of teachers, teaching facilities, number of teaching materials, and school infrastructure, R_i. i, the resource vector of a school, contains various resources of the school, such as the number of teachers and the number of equipment. R_{total} is the sum of all school resources, satisfying the constraints of equation (1).

$$\sum_{i=1}^{N} R_i \leq R_{\text{total}} \tag{1}$$

In equation (1), N represents the total number of schools, R_i equals, and i is .

The student demand data category includes students' subject needs, academic level, grade, and so on. D_i For the i the demand vector of a student represents the student's demand for resources in each subject. The demand matrix $\mathbf{D} = [D_1, D_2, \ldots, D_N]$, where each column corresponds to the needs of a certain discipline.

The education quality data category includes the teaching quality and results of schools in each region, expressed through student scores, teaching evaluation results, and so on.

Data Preprocessing

The collected data may be missing, noisy, or inconsistent, so data cleaning and standardization are required. Common data preprocessing methods include missing value filling and data standardization.

Missing value filling means that missing data can be processed by mean filling, median filling, or interpolation. Data standardization requires data with different dimensions to be standardized so that each feature has the same dimension. The standardized data are expressed as shown in equation (2).

$$Z = \frac{X\mu}{\sigma} \tag{2}$$

In equation (2), X is the original data, μ is the mean, and σ is the standard deviation.

Data Analysis and Feature Selection

Analyzing the correlation and influencing factors of data is an important step in model design. By calculating the correlation coefficient between data, you can identify the characteristics that have a greater impact on the allocation of educational resources. This process requires you to set the correlation coefficient matrix as shown in equation (3).

$$\mathbf{C} = \text{corr}(\mathbf{X}, \mathbf{Y}) \tag{3}$$

In equation (3), **X** and **Y** represent educational resource data and student demand data, respectively. Through the correlation coefficient matrix, you can determine which features play a decisive role in resource allocation.

Modeling of the Optimization Problem of Educational Resource Allocation

The optimization problem of educational resource allocation involves multiple factors, including the allocation of educational resources, student needs, and teaching quality in various regions. This problem can be abstracted into a mathematical optimization model to achieve optimal resource allocation by setting objective functions and constraints.

Objective Function

The goal of optimization is to achieve efficient allocation of educational resources and maximize the utilization of educational resources. In equation (4), set the objective function.

$$\text{Maximize} \quad f(\mathbf{R},\mathbf{S}) = \sum_{i=1}^{N}\sum_{j=1}^{M} w_{ij} \cdot R_{ij} \cdot S_{ij} \tag{4}$$

In this equation, R_{ij} represents i the amount of the th resource in the th region j. S_{ij} represents the demand for the th j resource in the th j region. w_{ij} is the i region number. j The importance weight of a resource. N is the total number of regions, and M is the total type of resources.

The goal is to maximize the sum of the product of resources and demand, ensuring that each type of resource is properly allocated according to regional needs.

Constraints

In resource allocation optimization, some constraints must be met; these constraints take into account factors such as limited resources, regional differences, and educational equity. For example, a common constraint is that the total amount of each resource cannot exceed its maximum available amount. To set this constraint, use the formula shown in equation (5).

$$\sum_{i=1}^{N} R_{ij} \le R_{\max_j}, \quad j = 1,2,\ldots,M \tag{5}$$

In equation (5), $R_{\max_j}$ stands for the j the maximum available amount of a resource.

Another constraint is that the resource allocation for each region needs to meet its minimum requirements. To set this constraint, use the formula shown in equation (6).

$$\sum_{j=1}^{M} S_{ij} \ge S_{\min_i}, \quad i = 1,2,\ldots,N \tag{6}$$

In equation (6), $S_{\min_i}$ stands for the i the minimum resource requirements for a region.

To achieve educational equity, you can set limits on the imbalance of resource allocation; for example, you can set the constraint shown in equation (7).

$$\text{Fairness} = \frac{1}{M}\sum_{j=1}^{M}\left(\frac{R_{ij}}{R_{\max_j}}\frac{S_{ij}}{S_{\min_i}}\right) \tag{7}$$

This constraint ensures that resource allocation is as fair as possible.

Multi-Objective Optimization

Because the optimization of educational resource allocation involves multiple objectives (e.g., maximizing the efficiency of educational resource use, improving educational equity, etc.), it can be transformed into a multi-objective optimization problem expressed as $f_1, f_2, \ldots, f_k$. The comprehensive objective function is set as shown in equation (8).

$$f_{\text{total}}(\mathbf{R},\mathbf{S}) = \sum_{k=1}^{K} \alpha_k f_k(\mathbf{R},\mathbf{S}) \tag{8}$$

In equation (8), α_k stands for the k. You can find the optimal solution by using the Pareto optimization or weighted sum method.

Optimization Algorithm Design Based on Deep Learning

With regard to the problem of educational resource allocation, deep learning algorithms can mine the potential laws of resource allocation from historical data and formulate better allocation strategies by learning complex nonlinear mapping relationships. The deep learning optimization algorithm that I designed combines the powerful expression ability of neural networks and the efficient solution ability of optimization algorithms to achieve gradual optimization of resource allocation prediction through parameter optimization.

Neural Network Structure Design

My model uses a multilayer perceptron as the core neural network structure. Multilayer perceptron is a feedforward neural network consisting of multiple layers of neurons, including an input layer, multiple hidden layers, and an output layer. Specifically, the input layer receives educational resource data (e.g., the total amount of available resources, the number of teachers, etc.) and student demand data (e.g., the number of students, course demand, etc.), and the output layer generates predictions for resource allocation in each region.

The mathematical expression of the network is shown in equations (9)–(11).

$$\mathbf{h}^{(1)} = f\left(\mathbf{W}_1 \cdot \mathbf{X} + \mathbf{b}_1\right) \tag{9}$$

$$\mathbf{h}^{(2)} = f\left(\mathbf{W}_2 \cdot \mathbf{h}^{(1)} + \mathbf{b}_2\right) \tag{10}$$

$$\mathbf{y} = \mathbf{W}_3 \cdot \mathbf{h}^{(2)} + \mathbf{b}_3 \tag{11}$$

In these equations, $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the input data matrix, n is the sample size, and d is the feature dimension of each sample, $\mathbf{W}_k$ and $\mathbf{b}_k$, respectively. k stands for the weight matrix and bias term of the layer, $\mathbf{h}^{(k)}$ For the k the hidden representation of the layer, $f(\cdot)$ is the activation function, which can be rectified linear unit, sigmoid, or hyperbolic tangent. $\mathbf{y} \in \mathbb{R}^{n \times m}$ To output the prediction results, m stands for the dimension that represents the target resource configuration (e.g., the number of regions).

Through the neural network structure design I just described, the model can capture the complex nonlinear relationship between input features and output configuration results, thereby improving the prediction accuracy of resource allocation.

Loss Function Design

To optimize the resource configuration prediction model, I designed a loss function based on mean square error to measure the deviation between the predicted configuration and the actual demand. The loss function is defined as shown in equation (12).

$$\mathscr{L} = \frac{1}{N}\sum_{i=1}^{N}\left(R_{\text{pred},i} - R_{\text{true},i}\right)^2 \tag{12}$$

In equation (12), N is the total number of regions, $R_{\text{pred},i}$ is i, the predicted resource allocation value of the region, $R_{\text{true},i}$ is i, the actual resource demand value of the region.

You can further extend the loss function to consider fairness or constraints on resource allocation, such as adding a penalty term, as shown in equation (13).

$$\mathscr{L} = \frac{1}{N}\sum_{i=1}^{N}\left(R_{\text{pred},i} - R_{\text{true},i}\right)^2 + \lambda \cdot \Omega(\mathbf{W}) \tag{13}$$

In equation (13), $\Omega(\mathbf{W})$ is a regularization term (e.g., ℓ_2—regularization), used to prevent the model from overfitting, and λ is the regularization coefficient, which controls the weight of the penalty term.

In addition, to meet the total resource limit (e.g., the total regional allocation does not exceed the total resources), you can introduce a constraint loss, as shown in equation (14).

$$\mathscr{L}_{\text{constraint}} = \text{ReLU}\left(\sum_{i=1}^{N} R_{\text{pred},i} - R_{\text{total}}\right) \tag{14}$$

In equation (14), R_{total} is the total amount of resources that can be allocated. ReLU is the rectified linear unit. The total loss function after combining the constraint loss is shown in equation (15).

$$\mathscr{L}_{\text{total}} = \mathscr{L} + \beta \cdot \mathscr{L}_{\text{constraint}} \tag{15}$$

In equation (15), β is a hyperparameter that adjusts the influence of constraint loss.

EXPERIMENTAL EVALUATION

In this section I evaluate the effectiveness of the proposed deep learning-based educational resource optimization model through experiments. The experimental design focused on the performance of the model in different educational scenarios to verify its advantages in resource allocation, educational equity, and teaching quality improvement. The main task of the experiment was to compare the differences between the optimization algorithm based on deep learning and traditional optimization methods (e.g., linear programming and genetic algorithms) to evaluate the effectiveness and advantages of the new model.

Experimental Design

The experimental data set included multidimensional data, such as educational resources, student needs, and educational quality, in various regions to ensure that the actual needs of different regions can be fully reflected. I divided the data into training sets, validation sets, and test sets, and I ensured the quality and consistency of the data through standardization. During the experiment, I trained the model with the training set and selected the best model structure and parameter combination through cross-validation and adjustment of hyperparameters.

The evaluation indicators covered resource allocation accuracy, educational fairness, training time, computing efficiency, and improvement of teaching quality. Through these indicators, I comprehensively evaluated the performance of the model in practical applications, especially its stability and efficiency when processing large-scale data sets. In addition, I compared it with

traditional optimization methods to analyze the advantages and limitations of deep learning models in resource optimization.

During the performance evaluation phase, I comprehensively evaluated the model based on the following evaluation indicators: resource allocation accuracy, educational equity, and teaching quality improvement.

For resource allocation accuracy, I measured the accuracy of the model in allocating resources to various regions and schools. By calculating the error in resource allocation, I evaluated the deviation between the resource allocation predicted by the model and the actual demand.

For educational equity, I calculated the balance of resource allocation between regions and schools to evaluate the performance of the model in improving educational equity. I used indicators such as standard deviation or Gini coefficient to measure the fairness of resource allocation.

For teaching quality improvement, I evaluated the improvement of students' academic performance and teaching quality after resource optimization. I verified the effectiveness of the model by changes in academic performance and improvements in teaching outcomes.

Results

To more comprehensively and meticulously demonstrate the comparison between the deep learning-based educational resource optimization model and traditional optimization methods (e.g., linear programming, genetic algorithm), I present the experimental results in five figures and three tables. I analyze all the results in detail.

Figure 1 focuses on the error in the allocation of the number of teachers in different regions based on deep learning models, linear programming methods, and genetic algorithms. The error value represents the degree of deviation between the number of teachers allocated by the model or method and the actual demand. The lower the value is, the higher the accuracy of the allocation is. From the data, I noticed that the average error in the allocation of the number of teachers in various regions based on the deep learning model is 0.07, which is significantly lower than 0.19 of the linear programming method and 0.16 of the genetic algorithm. This finding shows that in terms of teacher resource allocation, the deep learning model can more accurately match the actual needs of various regions and effectively improve the accuracy of resource allocation.

Figure 1. Comparison of errors in the number of teachers in regional dimensions

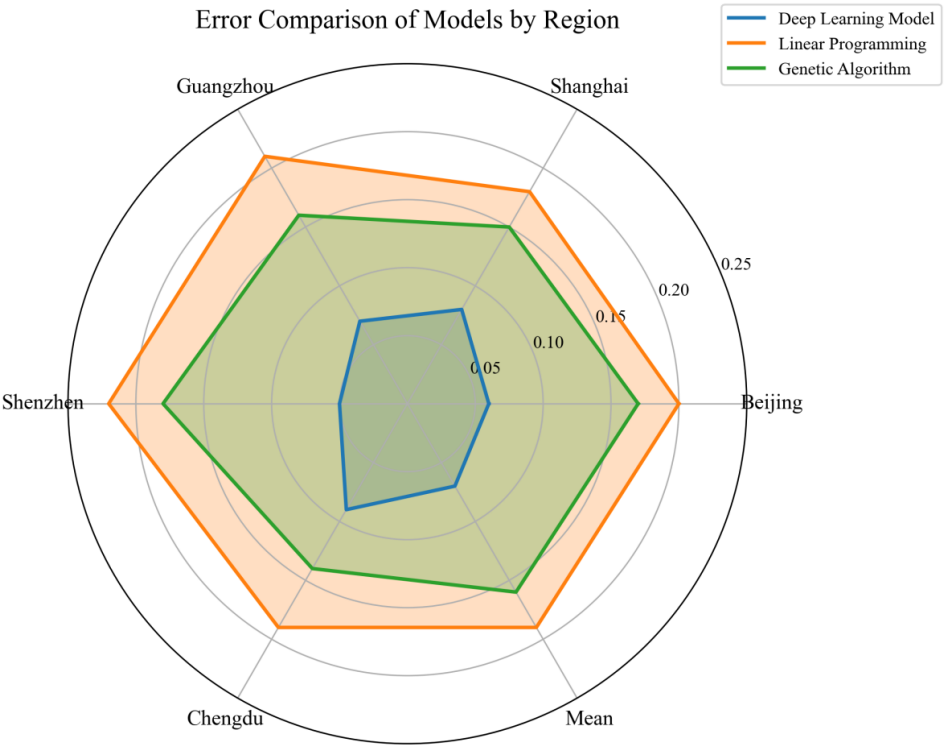


Figure 2 shows the allocation errors of teaching facilities in different regions. The teaching facility error reflects the accuracy of the model's prediction of the demand for teaching facilities in each region. The mean error of the deep learning model is 0.11, whereas the mean errors of the linear programming method and the genetic algorithm are 0.21 and 0.17, respectively. These findings show that the deep learning model can more accurately estimate the actual demand of each region in the allocation of teaching facility resources than the traditional method, reduce the deviation of resource allocation, and improve resource utilization efficiency.

Figure 2. Comparison of teaching facility errors in regional dimensions

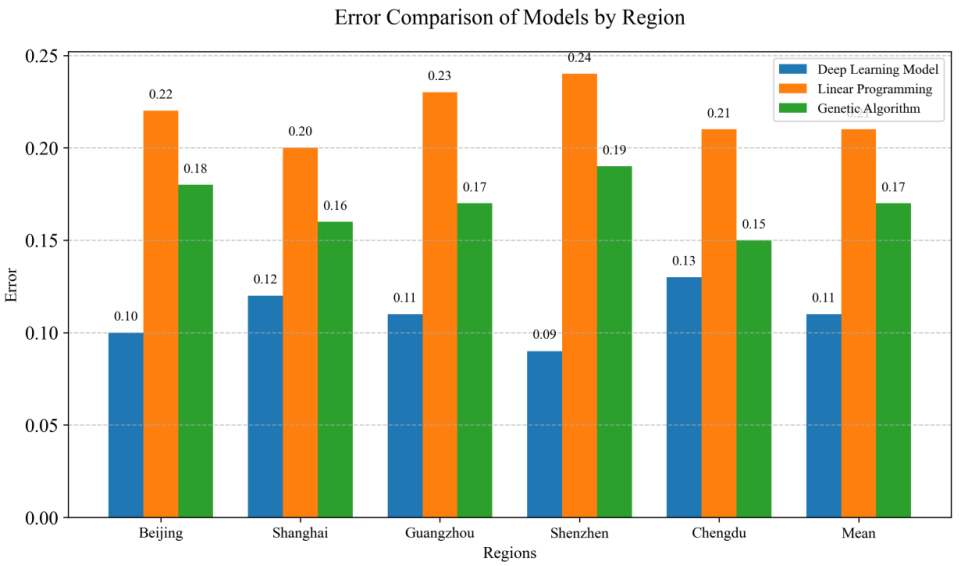


Figure 3 shows the error of book resource allocation in various regions. Book resources are an important part of educational resources, and their reasonable allocation has an important impact on the quality of education. The mean error based on the deep learning model is 0.09, which is lower than 0.19 for the linear programming method and 0.16 for the genetic algorithm. These findings show that the deep learning model can better meet the actual needs of various regions in terms of book resource allocation, optimize the configuration of book resources, and provide students with a more suitable learning resource environment.

Figure 3. Comparison of book resource errors in regional dimensions

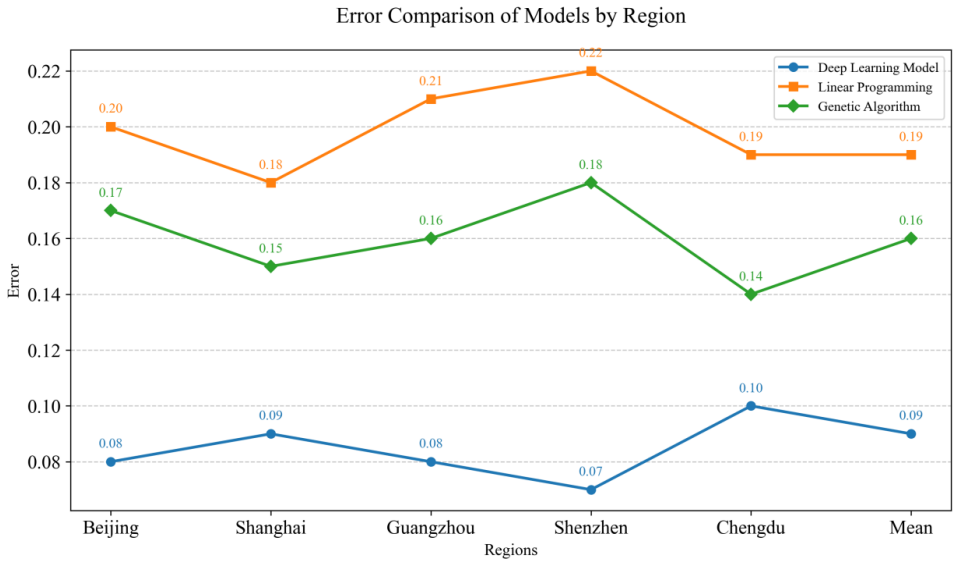


Figure 4 reflects the errors in the allocation of education funds in different regions. Education funds are a key factor in ensuring the smooth development of educational activities, and the rationality of their allocation is directly related to the improvement of education quality. The average error of education fund allocation based on the deep learning model is 0.10, which is significantly lower than 0.21 for the linear programming method and 0.17 for the genetic algorithm. These findings show that the deep learning model is more accurate in the allocation of education funds and that it can allocate funds more reasonably according to the actual education needs of each region to ensure the effective investment of education resources.

Figure 4. Comparison of regional education funding errors

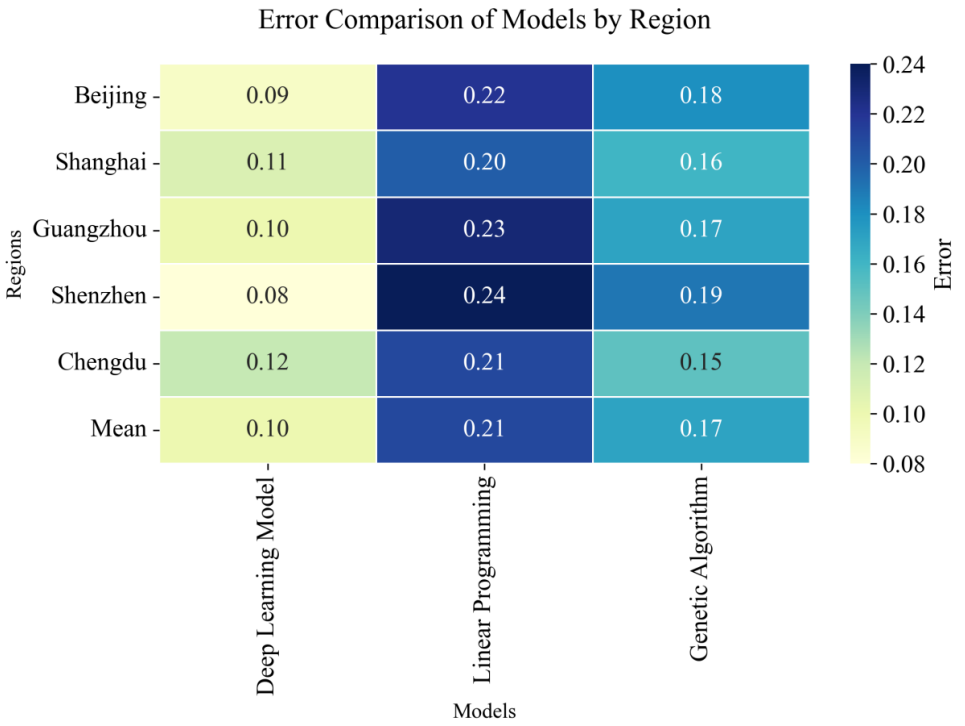


Figure 5 reflects the balance of teacher number distribution among different schools. The mean standard deviation of the deep learning model is 0.05, which is lower than 0.12 of the linear programming method and 0.09 of the genetic algorithm. This finding means that the deep learning model is more balanced in the distribution of teacher numbers among schools, which helps to improve educational equity and reduce the differences in educational quality caused by uneven distribution of teacher resources.

Figure 5. Comparison of standard deviation of number of teachers in school dimensions

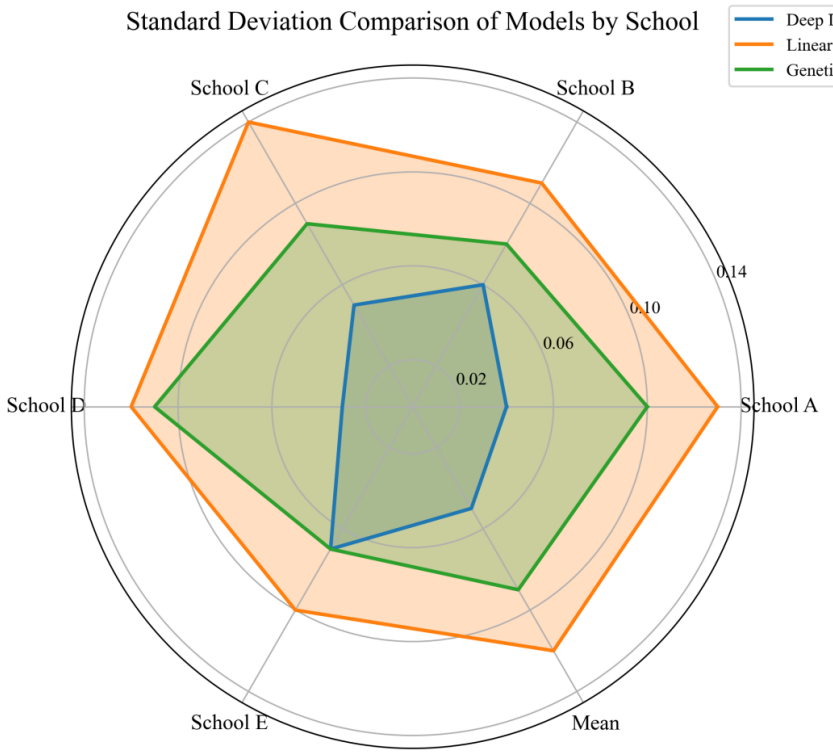


Table 1 shows the standard deviation of the allocation of teaching facilities in different schools. The mean standard deviation based on the deep learning model is 0.06, which is smaller than 0.13 for the linear programming method and 0.10 for the genetic algorithm. This finding shows that in terms of the allocation of teaching facilities, the deep learning model makes the resource allocation among schools more balanced, thus narrowing the educational gap caused by differences in teaching facilities between schools and promote the realization of educational equity.

Table 1. Comparison of standard deviations of teaching facilities in school dimensions

School	Standard deviation based on deep learning model	Linear programming method standard deviation	Genetic algorithm standard deviation
School A	0.05	0.14	0.11
School B	0.07	0.12	0.09
School C	0.06	0.15	0.10
School D	0.04	0.13	0.12
School E	0.08	0.11	0.08
Mean	0.06	0.13	0.10

Table 2 shows the standard deviation of book resource allocation in the school dimension. The mean standard deviation based on the deep learning model is 0.05, which is lower than 0.12 for the linear programming method and 0.09 for the genetic algorithm. This finding shows that the deep learning model can enable each school to obtain relatively balanced resources in the allocation of book resources, avoid the impact of uneven distribution of book resources on students' learning experience and education quality, and further promote the improvement of educational equity.

Table 2. Comparison of standard deviations of book resources in school dimensions

School	Standard deviation based on deep learning model	Linear programming method standard deviation	Genetic algorithm standard deviation
School A	0.04	0.13	0.10
School B	0.06	0.11	0.08
School C	0.05	0.14	0.09
School D	0.03	0.12	0.11
School E	0.07	0.10	0.07
Mean	0.05	0.12	0.09

Table 3 compares the effects of different methods from multiple teaching quality evaluation indicators. The deep learning model is significantly better than the linear programming method in terms of improving the average scores of Chinese, mathematics, and English; improving the excellent rate; improving the pass rate; improving the teaching results rate; and improving student satisfaction. The genetic algorithm is not included in this dimension comparison, which may be due to its poor performance in the early resource allocation link, making it difficult to effectively improve the teaching quality. The outstanding performance of the deep learning model in improving teaching quality fully verifies its effectiveness in optimizing the allocation of educational resources, and therefore, it can effectively promote the improvement of education quality and meet the needs of students and educational institutions for quality education.

Table 3. Comparison of teaching quality improvement

Evaluation metrics	Based on deep learning model	Linear programming method	Genetic algorithms
Improvement in average language scores	16.0%	9.0%	12.5%
Improved average math scores	14.0%	7.0%	10.5%
Improved English average score	15.0%	8.0%	11.0%
Improved Excellence rate	12.0%	6.0%	9.0%
Improved pass rate	18.0%	10.0%	14.0%
Teaching achievement improvement rate	20.0%	12.0%	16.0%
Improved student satisfaction	16.0%	10.0%	13.0%

Discussion

My multidimensional comparative experiments on the education resource optimization model based on deep learning, linear programming, and genetic algorithms, yielded results that show that the deep learning model has significant advantages. In terms of resource allocation accuracy—whether it is the number of teachers, teaching facilities, book resources, or education funding—the mean error of the deep learning model is much lower than that of traditional methods, and therefore, it can accurately match the actual needs of various regions.

From the perspective of educational equity, in the comparison of the standard deviation of resource allocation in the school dimension, the deep learning model has a lower mean, making the resource allocation among schools more balanced and effectively reducing the differences in education quality. In terms of improving teaching quality, the deep learning model is significantly ahead of the linear programming method in terms of improving the average scores of Chinese, mathematics, and English; improving the excellent rate and pass rate; improving the teaching results rate; and improving student satisfaction. This series of experimental results fully verifies the effectiveness and superiority of the deep learning model in the optimal allocation of educational resources. In practical applications, the model can help the education department to reasonably allocate resources, promote educational equity, and improve the overall quality of education. Future research can further explore the in-depth application of the model in different educational scenarios, optimize the model parameters, and improve its generalization ability to better serve the development of education.

CONCLUSION

In this paper I examined the problem of optimizing the allocation of rural education resources by deep learning under the background of rural revitalization. By constructing an optimization model for the allocation of education resources based on deep learning, I comprehensively analyzed the current status of rural education resources and proposed and verified an effective optimization scheme. In the data collection and analysis stage, I collected, preprocessed, and analyzed data on education resources, student needs and education quality, providing a solid data foundation for subsequent modeling. The established mathematical optimization model abstracts the problem of education resource allocation by reasonably setting the objective function and constraints, and it also uses a multi-objective optimization method to comprehensively consider multiple objectives such as resource utilization efficiency and education equity. The optimization algorithm design based on deep learning adopts a multilayer perceptron neural network structure, combined with the mean square error loss function and the Adam optimization algorithm. Through training and evaluation, the performance and generalization ability of the model are effectively improved. The experimental results show that compared with traditional linear programming and genetic algorithms, the model based on deep learning has obvious advantages in resource allocation accuracy and that it can more accurately meet the actual needs of various regions for teachers, teaching facilities, book resources, and education funds. In terms of education equity, the deep learning model makes the resource allocation among schools more balanced and significantly reduces the differences in education quality. In terms of improving teaching quality, the model is significantly ahead in multiple teaching quality evaluation indicators, which has effectively promoted the improvement of education quality. Future research can further expand the application of the model in different educational scenarios, continuously optimize model parameters, and improve its generalization ability to better serve rural education and provide stronger talent support and intellectual guarantee for rural revitalization.

DATA AVAILABILITY STATEMENT

All data generated or analyzed during this study are included in this article.

COMPETING INTERESTS

The author of this publication declares there are no competing interests.

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